RSNA - 脊柱病变识别银牌算法概览

竞赛概览：

本次竞赛要求参赛者通过算法建模解决一个重要的医疗挑战：如何利用腰椎磁共振成像（MRI）数据，辅助检测和分类退行性脊柱疾病。比赛使用了一个从全球八个机构收集的多中心、高质量数据集，包含腰椎 MRI 影像，以及五种退行性病变在五个椎间盘水平（L1/L2、L2/L3、L3/L4、L4/L5、L5/S1）的严重程度评分（正常/轻度、中度、重度）。参赛者的任务是模拟放射科医生的诊断过程，对左侧神经孔狭窄、右侧神经孔狭窄、左侧椎旁狭窄、右侧椎旁狭窄和椎管狭窄这五种退行性疾病进行精准分类。优秀参赛者的工作将有助于医疗机构开发出能够自动、准确、快速分类退行性腰椎疾病的工具，从而改进诊断的标准化，指导治疗和手术决策，最终提升患者的健康和生活质量。比赛的评估标准是加权的对数损失（log loss），根据不同严重程度的样本赋予不同的权重，以衡量模型在各个分类上的预测性能。

所用算法：

本次比赛利用 yolov8 识别和分类腰椎退化症状，算法流程：

1. 数据处理：根据提供的磁共振成像图片以及腰椎椎体的定点坐标位置信息构建目标检测数据，在多种放缩尺寸图像上设置合适大小的 box，最大化保留严重退化（Severe）特征信息。
2. 建模 1：将病例图像分成三大类：
   - spinal canal stenosis 椎管狭窄（15 类）
   - subarticular stenosis 椎间孔狭窄（30 类）
   - neural foraminal narrowing 神经根管狭窄（30 类）

   进行建立模型训练。在训练时对图像进行增强。
3. 建模 2：此方式与建模 1 类似，不同之处在于标签屏蔽。由于存在单个病例图像具有多个标签的情况，而且 yolov8 不具有多标签多分类功能，因此在数据处理阶段将具有多标签（同时存在严重退化与非严重退化）的图片去除非严重退化（即 normal_mild 和 moderate）类别，加强对严重退化类别的识别能力。
4. 推理：在每个病例的三大类图像中，分别采取对应的模型推理，在每个系列图像中分别得到所有类型的概率，取每种类别的最大值概率即为单个病例每种类别的概率，将所得概率归一化即为最后分类预测结果。

Silver Medal Algorithm Overview for RSNA 2024 Lumbar Spine Degenerative Classification

Competition Overview:

This competition requires participants to solve an important medical challenge through algorithm modeling: how to utilize lumbar spine magnetic resonance imaging (MRI) data to assist in the detection and classification of degenerative spine diseases. The competition uses a multicenter, high-quality dataset collected from eight institutions worldwide, containing lumbar MRI images and severity ratings (Normal/Mild, Moderate, Severe) for five types of degenerative lesions at five intervertebral disc levels (L1/L2, L2/L3, L3/L4, L4/L5, L5/S1). Participants are tasked with simulating the diagnostic process of radiologists to accurately classify five degenerative conditions: Left Neural Foraminal Narrowing, Right Neural Foraminal Narrowing, Left Subarticular Stenosis, Right Subarticular Stenosis, and Spinal Canal Stenosis. Outstanding participants' work will help medical institutions develop tools that can automatically, accurately, and quickly classify degenerative lumbar spine diseases. This advancement will improve diagnostic standardization, guide treatment and surgical decisions, and ultimately enhance patients' health and quality of life. The competition's evaluation metric is weighted logarithmic loss (log loss), assigning different weights to samples based on severity levels to measure the model's predictive performance across various classifications.

Algorithm Descriptions:

In this competition, we utilized YOLOv8 to recognize and classify lumbar spine degenerative symptoms. The algorithm workflow is as follows:

1. Data Processing: Based on the provided MRI images and the precise coordinate information of the lumbar vertebrae, we constructed target detection data. Appropriate bounding boxes were set on images of various scales to maximize the retention of severe degenerative features.
2. Modeling 1: We divided the case images into three major categories for model training:
   - Spinal Canal Stenosis (15 classes)
   - Subarticular Stenosis (30 classes)
   - Neural Foraminal Narrowing (30 classes)

   During training, we applied data augmentation techniques to enhance the images.

3. Modeling 2: This approach is similar to Modeling 1 but differs in label masking. Since some case images have multiple labels and YOLOv8 does not support multi-label multi-classification, we removed non-severe degeneration categories (i.e., Normal_Mild and Moderate) from images with multiple labels during the data processing stage. This step was taken to strengthen the model's ability to recognize severe degeneration categories.
4. Inference: For each of the three categories of images in each case, we used the corresponding models for inference. In each image series, we obtained probabilities for all types and selected the maximum probability for each category as the probability for that category in a single case. The obtained probabilities were then normalized to produce the final classification prediction results.